

Problem 17

Evaluate the integral:

$$\int \sqrt{1 - \sin 2x} \, dx.$$

Solution

We are tasked with solving the integral:

$$\int \sqrt{1 - \sin 2x} \, dx.$$

Step 1: Simplify $\sin 2x$ using a trigonometric identity

Using the double-angle identity for sine:

$$\sin 2x = 2 \sin x \cos x,$$

we substitute this expression into the integral. The integral becomes:

$$\int \sqrt{1 - 2 \sin x \cos x} \, dx.$$

To simplify the expression further, we use the identity:

$$1 - \sin 2x = 1 - 2 \sin x \cos x = (\cos x - \sin x)^2.$$

Thus, the integral becomes:

$$\int \sqrt{1 - \sin 2x} \, dx = \int \sqrt{(\cos x - \sin x)^2} \, dx.$$

Step 2: Simplify the square root

The square root of $(\cos x - \sin x)^2$ is $|\cos x - \sin x|$. For simplicity, we consider the principal value:

$$\sqrt{(\cos x - \sin x)^2} = \cos x - \sin x.$$

The integral now becomes:

$$\int \sqrt{1 - \sin 2x} \, dx = \int (\cos x - \sin x) \, dx.$$

Step 3: Evaluate the integral

We split the integral into two simpler parts:

$$\int (\cos x - \sin x) \, dx = \int \cos x \, dx - \int \sin x \, dx.$$

1. **First term: $\int \cos x \, dx$:** The integral of $\cos x$ is:

$$\int \cos x \, dx = \sin x.$$

2. **Second term: $\int \sin x \, dx$:** The integral of $\sin x$ is:

$$\int \sin x \, dx = -\cos x.$$

Combining these results:

$$\int (\cos x - \sin x) \, dx = \sin x + \cos x.$$

Step 4: Add the constant of integration

Finally, the solution to the integral is:

$$\int \sqrt{1 - \sin 2x} \, dx = \sin x + \cos x + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$\int \sqrt{1 - \sin 2x} \, dx = \sin x + \cos x + C.$$